

hypothesis based on some sort of structural theory or qualitative understanding of the market, and then test that hypothesis to see whether it is supported by the data.

[Woodruff speaking emphatically] *I don't do that.* I read all of that just to get to the point that I do what I am not supposed to do, which is a really interesting observation because I am supposed to fail. According to almost everyone, you have to approach systematic trading (and predictive modeling in general) from the framework of “Here is a valid hypothesis that makes sense within the context of the markets.” Instead, I blindly search through the data.

It's nice that people want hypotheses that make sense. But I thought that was very limiting. I want to be able to search the rest of the stuff. I want to automate that process. If you set the problem up really well with cross validation, then overfitting is a problem that can be overcome. I hypothesized that there are patterns that work, and I would rather have the computer test trillions of patterns than just a few hundred that I had thought of.

There is one aspect of the process, though, that is manual. The secondary variables that are used to construct price-forecasting models have to make sense. For example, it makes sense that price-derived data series, such as volatility or price acceleration, might provide important information. The list of secondary variables derived from price is the part I built manually. Then I have a framework for combining the secondary variables in all sorts of combinations to see what works.

I wanted to hand that work off to the computer, but I knew how important it was to have the hindsight bias and overfit problem figured out. As an aside, I am still trying to reverse engineer some of the models that we have come up with that are so interesting and amazing. What do these patterns say about the psychology of the marketplace? Frankly, I'm not sure yet.

**You are constructing models by selecting combinations of secondary variables formed from a list of hundreds of possible secondary variables. Depending on the specific selection constraints you use, there would be millions if not billions of possible combinations.**